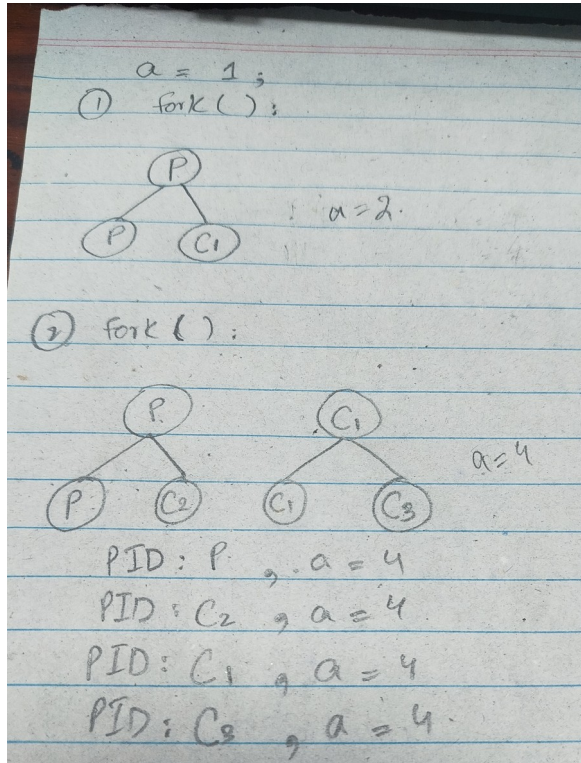


LAB 4 Tasks

Q1.

1. 4 processes
2. 4 times
3. All 4
- 4.



5. Output can not be predicted as it depends on the scheduling of OS that whether it executes parent process or child process first.

Q2.

1. `exec()` replaces the current process with the new process from the given path and starts its execution. In this program, Hello from exec! Will be displayed
2. Before replacing the process
Hello from exec!

These lines will execute.

3. It will not execute as `exec()` is successful and it replaces the current process with new process given through the path.
4. If the path was incorrect, the `exec()` would fail and the current process execution will continue, printing the statement: "After exec-will this line run"
5. If we remove `exec()` system call, both `printf` lines will be displayed.

Q3.

With `wait()`, the parent waits for the child process execution then continues its execution.

Without `wait()` statement, the parent process does not wait for completion of child process and continues its execution. After parent process is finished, the child process is finished as the child process sleeps and then continues so "Child finished" is displayed in the end.